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Germain et al.

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(54) **TISSUE RESECTING DEVICE AND METHODS**

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A61B 18/14 (2006.01)
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(Continued)

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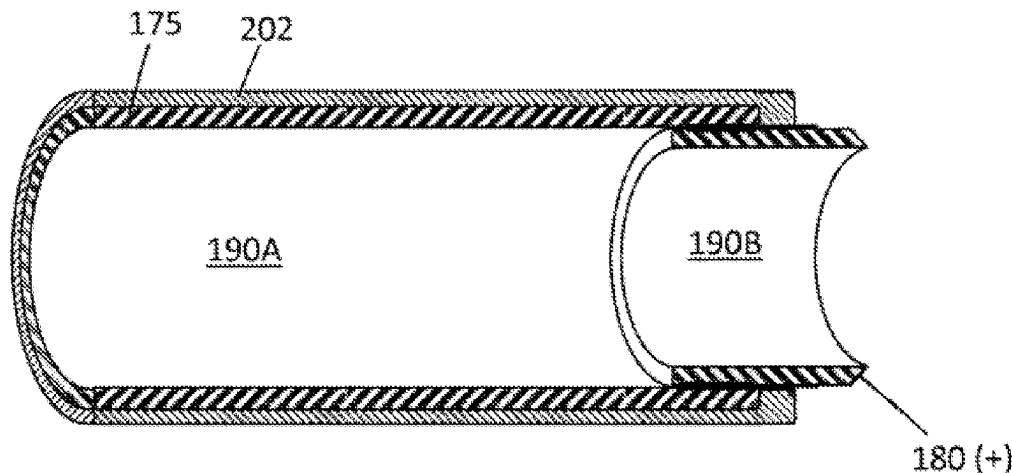
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(57) **ABSTRACT**

Devices, systems, and methods for resecting tissue are disclosed. In some embodiments, a tissue resecting device may comprise an elongated structure having a longitudinal axis, the elongated structure comprising an outer sleeve with a distal window configured to receive uterine polyp tissue and an inner sleeve configured to move between a proximal position and a distal position relative to the window. In some further embodiments, the device may also comprise an electrode element coupled to the inner sleeve. In some even further embodiments, the device may include an insulative layer covering at least a portion of the inner sleeve, wherein the tissue resecting device is configured to fail when used to resect tissue more fibrous than uterine polyp tissue.

20 Claims, 10 Drawing Sheets



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- continuation of application No. 15/245,586, filed on Aug. 24, 2016, now abandoned.
- (60) Provisional application No. 62/210,540, filed on Aug. 27, 2015.

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(52) U.S. Cl.

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See application file for complete search history.

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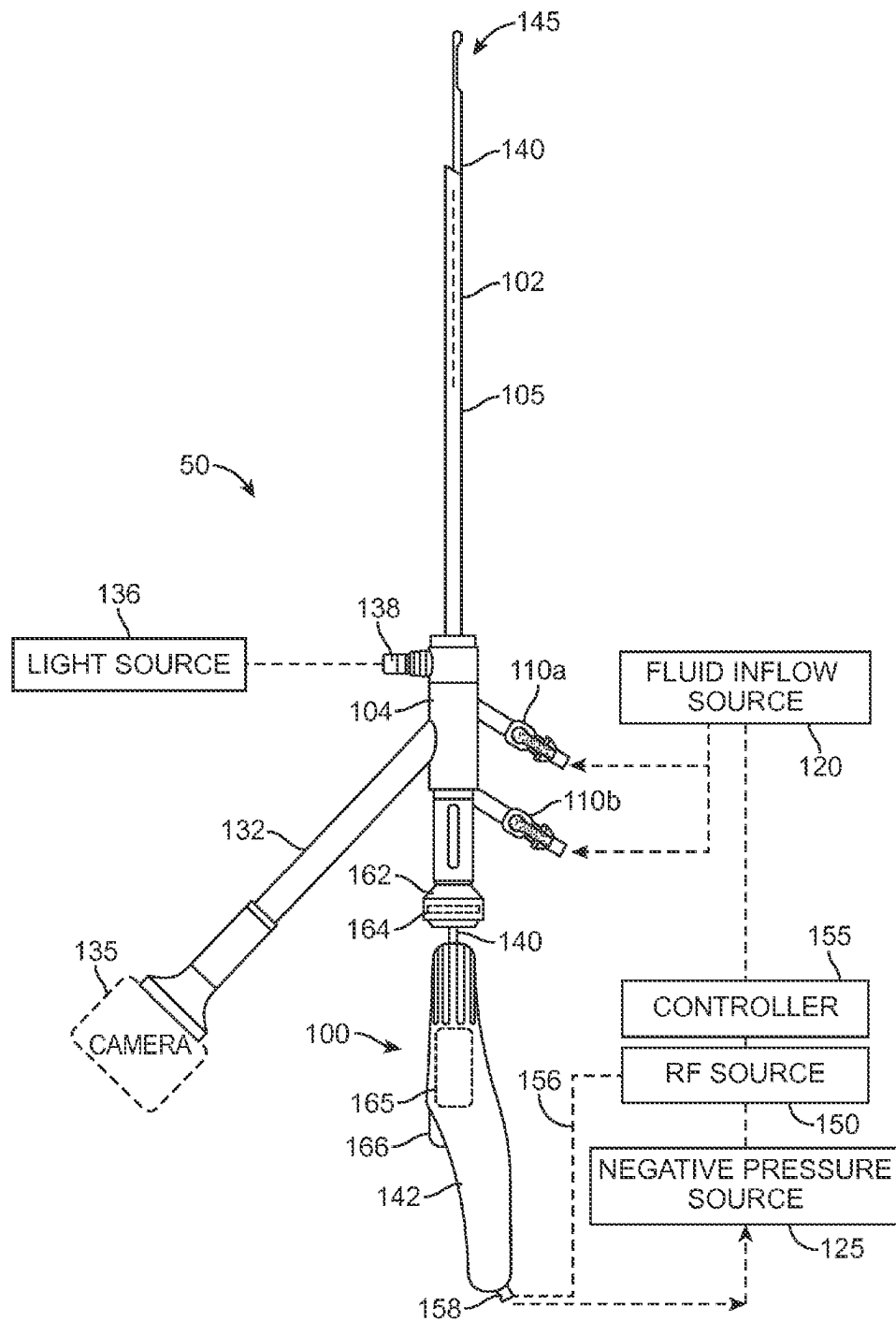


FIG. 1

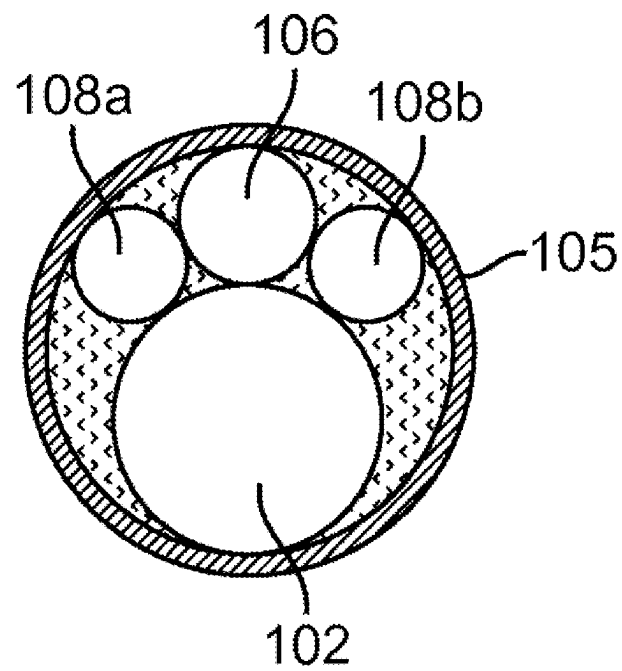
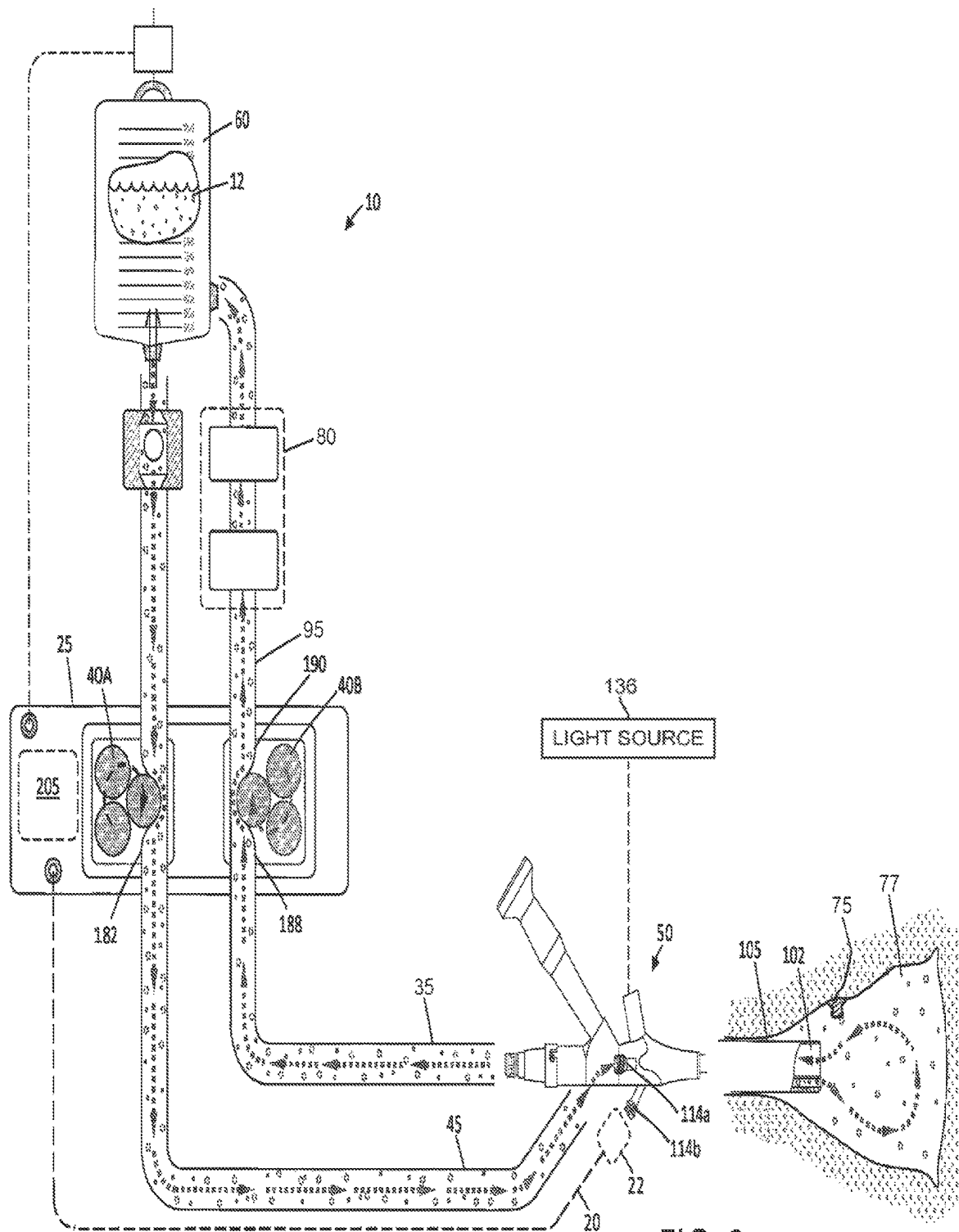
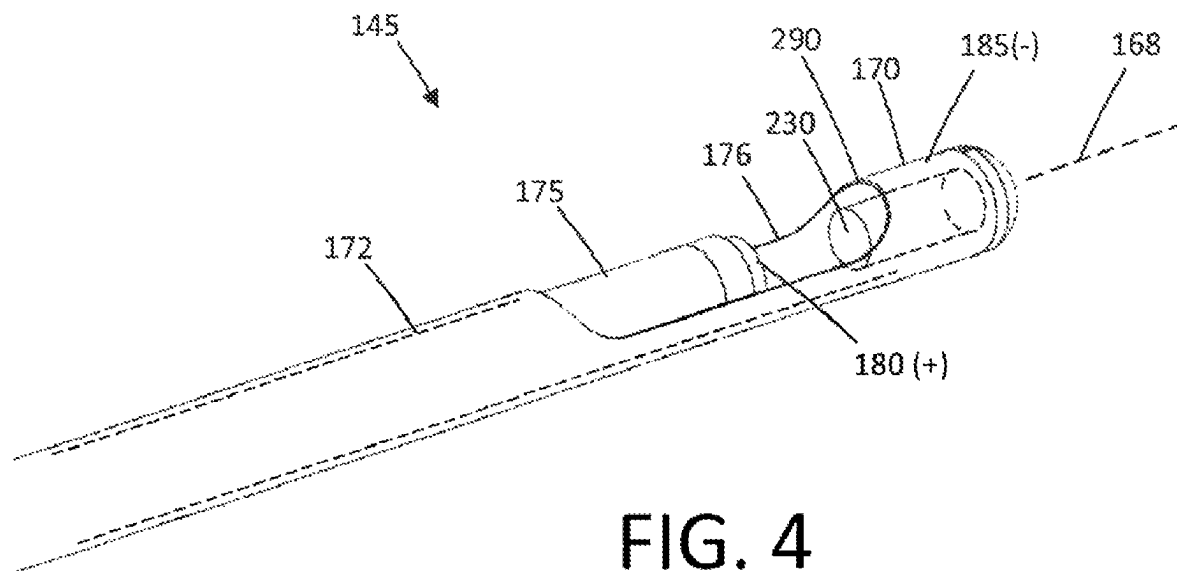


FIG. 2





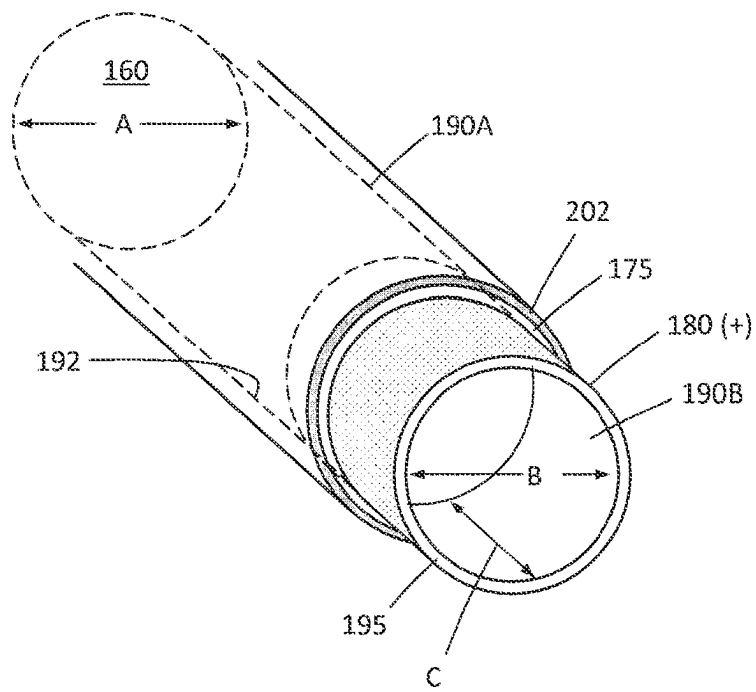


FIG. 5

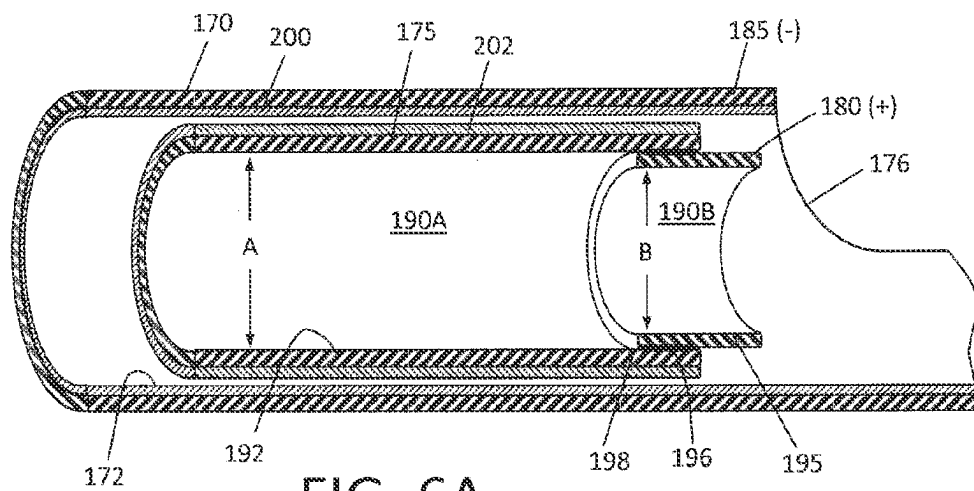


FIG. 6A

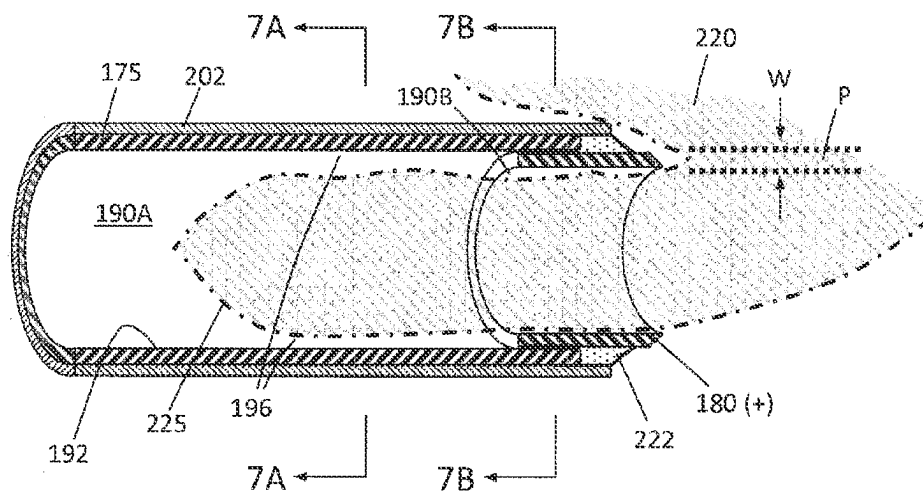


FIG. 6B

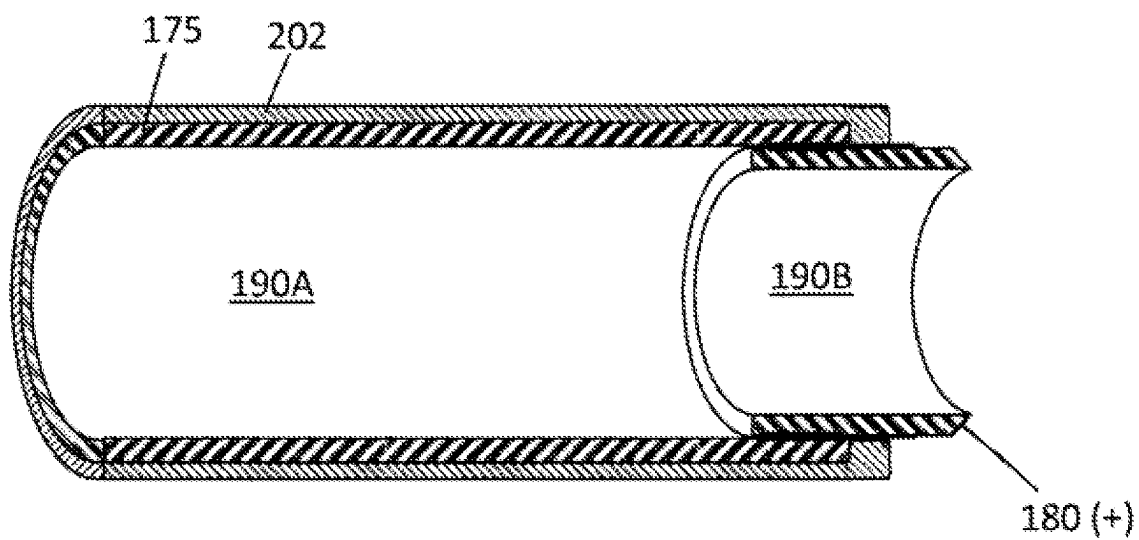


FIG. 7A

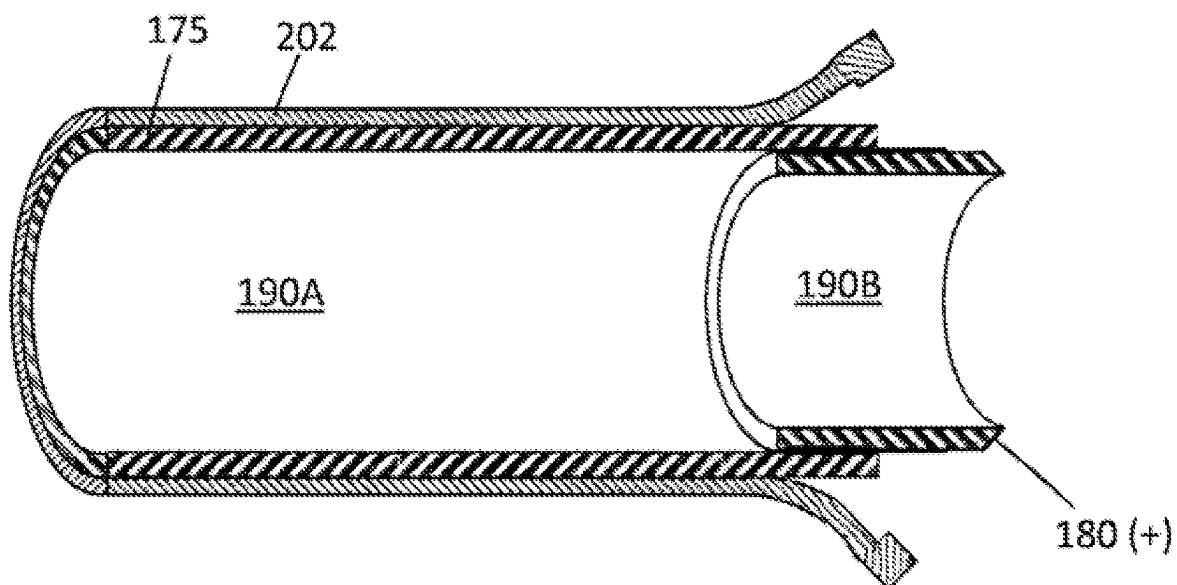


FIG. 7B

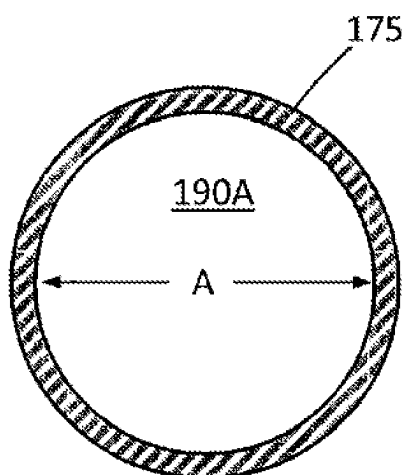


FIG. 8A

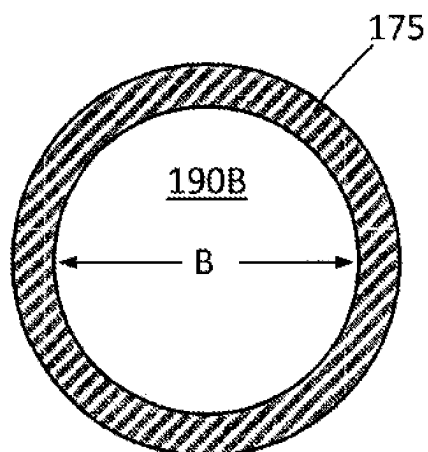


FIG. 8B

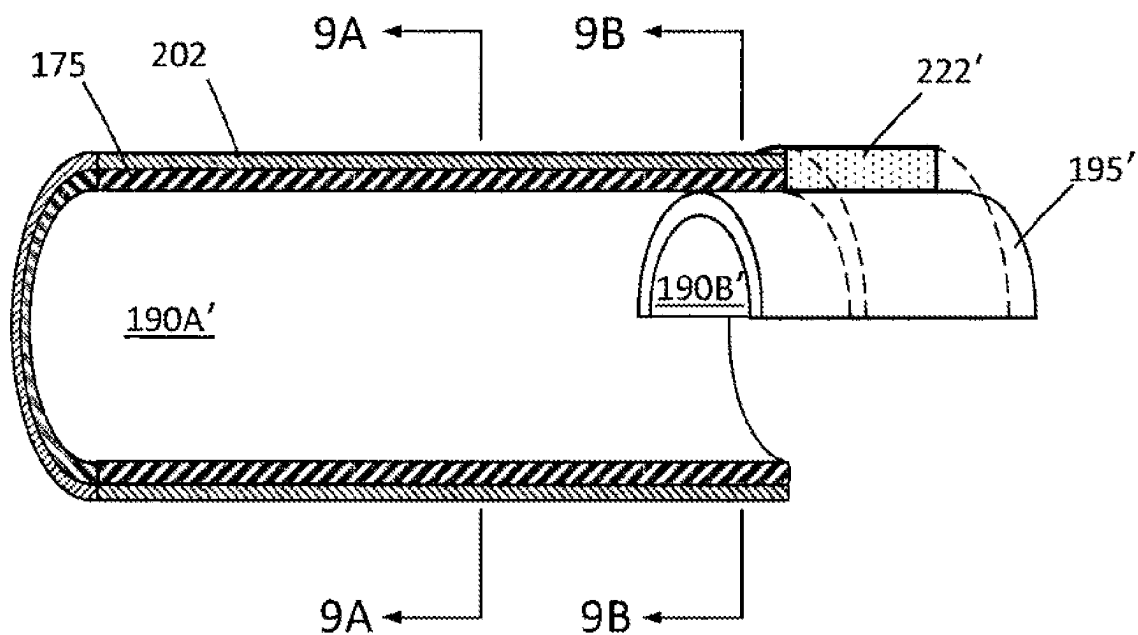


FIG. 9

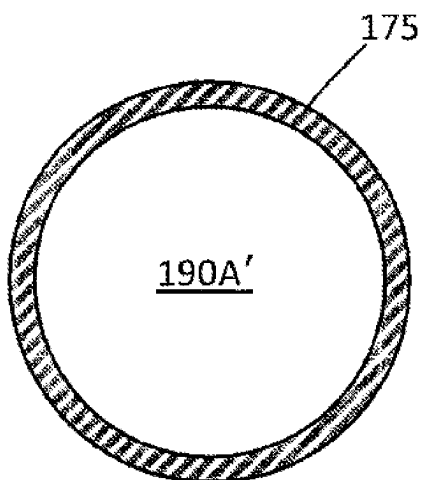


FIG. 10A

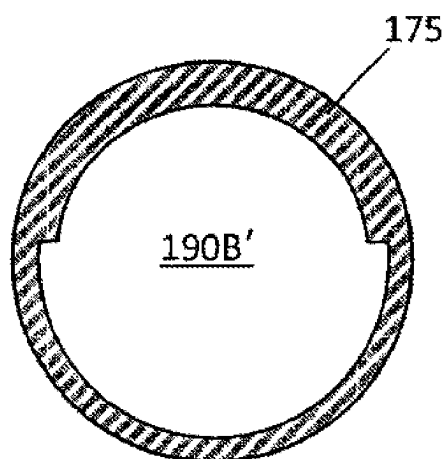


FIG. 10B

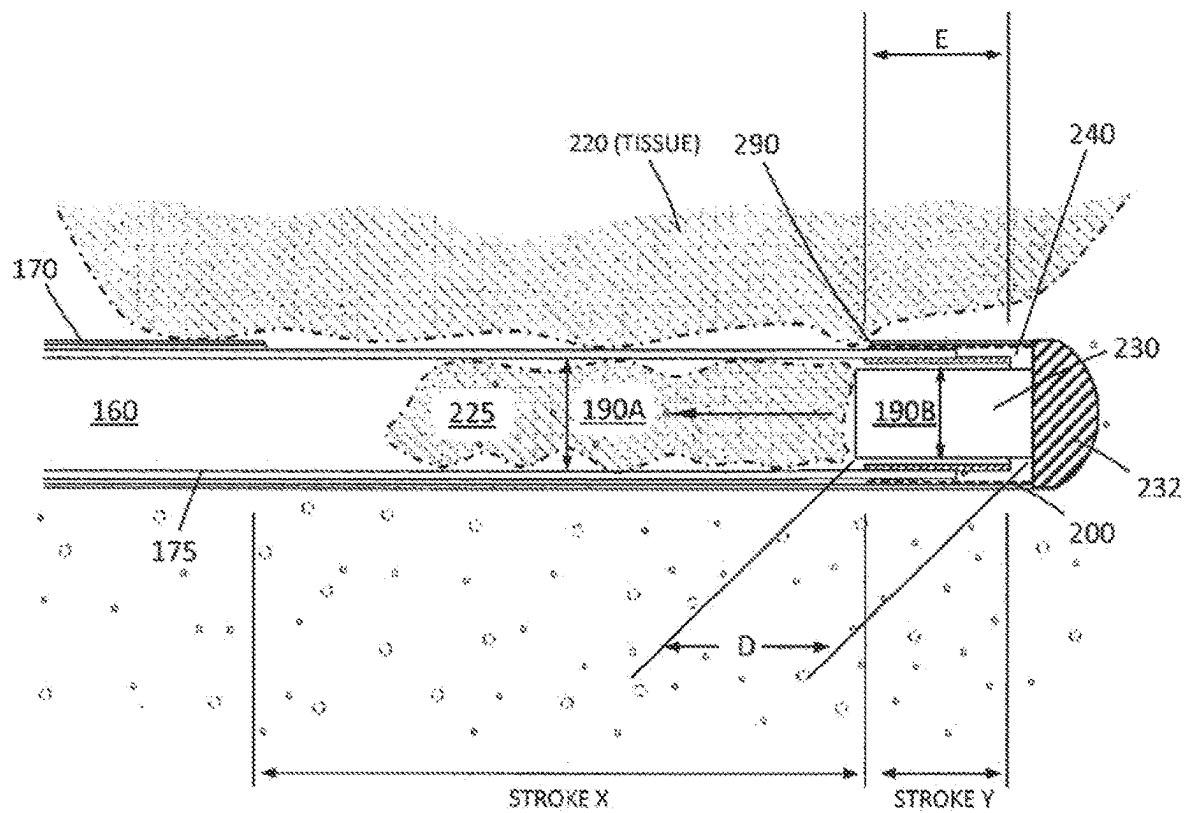


FIG. 11

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TISSUE RESECTING DEVICE AND METHODS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/500,156, filed Oct. 13, 2021, now U.S. Pat. No. 11,957,406, which is a continuation of U.S. application Ser. No. 15/245,586, filed Aug. 24, 2016, now abandoned, which application claims priority to U.S. Provisional Patent Application Ser. No. 62/210,540, filed on Aug. 27, 2015, the contents of which are fully incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates systems and methods for the resection and extraction of tissue, for example, uterine polyps and other abnormal uterine tissue.

BACKGROUND

Uterine polyps are growths attached to the inner wall of the uterus that extend into the uterine cavity. Uterine polyps are usually non-cancerous and can range in size from a few millimeters to a few centimeters. Uterine polyps may cause menorrhagia, bleeding between menstrual periods, reproductive dysfunction, pelvic pressure and pain.

One current treatment of polyps is hysteroscopic resection or myomectomy which involves transcervical access to the uterus with a hysteroscope together with insertion of a resecting instrument through a working channel in the hysteroscope. The resecting instrument may be an electro-surgical resection device such as an RF loop. An electro-surgical resecting device is disclosed in U.S. Pat. No. 5,906,615. In other instances, a mechanical cutter may be used to mechanically cut tissue. Mechanical cutting devices are disclosed in U.S. Pat. Nos. 7,226,459; 6,032,673; 5,730,752; and U.S. Published Patent Appl. 2009/0270898.

SUMMARY

The present disclosure relates systems and methods for resection and extraction of tissue, for example, uterine polyps and other abnormal uterine tissue.

In a first illustrative embodiment, a tissue resecting device for resecting uterine polyps may comprise an elongated structure having a longitudinal axis, the elongated structure comprising an outer sleeve with a distal window configured to receive uterine polyp tissue and an inner sleeve configured to move between a proximal position and a distal position relative to the window and an electrode element coupled to the inner sleeve. In at least some additional embodiments, the device may further include an insulative layer covering at least a portion of the inner sleeve and at least a portion of the electrode element, wherein the tissue resecting device is configured to fail when used to resect tissue more fibrous than uterine polyp tissue.

Another tissue resecting device for resecting uterine polyps comprises an elongated structure having a longitudinal axis. The elongated structure comprises an outer sleeve with a distal window configured to receive uterine polyp tissue and an inner sleeve configured to move between a proximal position and a distal position relative to the window. An electrode element having a first polarity is coupled to the inner sleeve and movable across the window between the proximal position and the distal position. An insulative layer

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is covering the inner sleeve proximal of the electrode element, wherein a portion of the insulative layer is exposed in the window in the distal position. The insulative layer is configured such that a degree of contact between the insulative layer and the inner sleeve is reduced when used to resect tissue more fibrous than uterine polyp tissue to expose a portion of the inner sleeve to alter an electrical pathway between the electrode element and the outer sleeve serving as a return electrode having a second polarity opposite the first polarity.

Additionally, or alternatively, in any of the above embodiments, the insulative material is configured to delaminate from the inner sleeve when used to resect tissue more fibrous than uterine polyp tissue.

Additionally, or alternatively, in any of the above embodiments, when the tissue resecting device fails, the insulative material may be configured to peel back from the inner sleeve.

Additionally, or alternatively, in any of the above embodiments, the insulative layer may be bonded directly to the electrode element.

Additionally, or alternatively, in any of the above embodiments, the distal window may have a longitudinal length of between about 5 mm and about 10 mm.

Additionally, or alternatively, in any of the above embodiments, the distal window may have a longitudinal length of about 8 mm.

Additionally, or alternatively, in any of the above embodiments, the insulative material may comprise fluorinated ethylenepropylene (FEP).

Additionally, or alternatively, in any of the above embodiments, the inner sleeve may comprise 304 stainless steel.

Additionally, or alternatively, in any of the above embodiments, the outer sleeve may comprise 304 stainless steel.

Additionally, or alternatively, in any of the above embodiments, the insulative layer may have a thickness of between about 0.0127 mm and about 0.038 mm.

Additionally, or alternatively, in any of the above embodiments, the insulative layer may be configured to detach from the electrode component after a predetermined period of activation of the tissue resecting device.

In another embodiment, a tissue resecting device may comprise an elongated assembly comprising concentric outer and inner sleeves extending along an axis, and the inner sleeve may further comprise an electrode element. In some additional embodiments, the device may further include a tissue-receiving window in the outer sleeve and insulative material disposed about at least a portion of the inner sleeve, wherein the insulative material is configured to detach from the inner sleeve when the tissue resecting device is used to resect tissue having a greater fibrosity than uterine polyp tissue.

Additionally, or alternatively, in any of the above embodiments, the electrode element may comprise at least a portion of the inner sleeve.

Additionally, or alternatively, in any of the above embodiments, the insulative material may be attached to the electrode element.

Additionally, or alternatively, in any of the above embodiments, the insulative material may comprise fluorinated ethylenepropylene (FEP).

Additionally, or alternatively, in any of the above embodiments, the insulative material may be configured to detach from the electrode element after a predetermined period of operation of the tissue resecting device.

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Additionally, or alternatively, in any of the above embodiments, the tissue-receiving window may have a longitudinal length of between about 5 mm and about 10 mm.

Additionally, or alternatively, in any of the above embodiments, the tissue-receiving window may have a longitudinal length of about 8 mm.

Additionally, or alternatively, in any of the above embodiments, the inner sleeve, the outer sleeve, and the tissue-receiving window may be configured for resecting uterine polyp tissue.

Additionally, or alternatively, in any of the above embodiments, the tissue resecting device may further comprise insulative material disposed along at least a portion of an inner surface of the outer sleeve.

In still another embodiment, a medical device system may comprise an elongated probe comprising: an outer sleeve and an inner sleeve, the outer sleeve including a window disposed proximate a distal end of the outer sleeve, an electrode element disposed proximate a distal end of the inner sleeve, and an insulative layer insulating the inner sleeve from the outer sleeve. In some embodiments, the insulative layer may be configured to peel away from the inner sleeve when the elongated probe is used to resect uterine fibroids. In some additional embodiments, the system may further include a motor for reciprocating the inner sleeve relative to the outer sleeve and an RF generator for delivering energy through the electrode element to resect tissue.

Additionally, or alternatively, in any of the above embodiments, the insulative layer may be attached to at least a portion of the electrode element.

Additionally, or alternatively, in any of the above embodiments, the window may have a longitudinal length of between about 5 mm and about 10 mm.

The above summary of some embodiments is not intended to describe each disclosed embodiment or every implementation of the present disclosure. The Figures, and Detailed Description, which follow, more particularly exemplify these embodiments.

BRIEF DESCRIPTION OF DRAWINGS

The disclosure may be more completely understood in consideration of the following detailed description of various embodiments in connection with the accompanying drawings, in which:

FIG. 1 is a plan view of an assembly including a hysteroscope and a tissue-resecting device inserted through the working channel of the hysteroscope;

FIG. 2 is a cross sectional view of the shaft of the hysteroscope of FIG. 1;

FIG. 3 is a schematic view of a fluid management system that re-circulates distention fluid illustrated in conjunction with a hysteroscope;

FIG. 4 is a schematic view of the working end of the tissue-resecting device of FIG. 1 showing an outer sleeve in conjunction with an inner sleeve in a partially advanced position;

FIG. 5 is a schematic perspective view of the working end of the inner sleeve of FIG. 4;

FIG. 6A is a schematic cut-away view of a portion of the outer sleeve and the inner sleeve of FIG. 4;

FIG. 6B is another schematic cut-away view of a portion of the outer sleeve and the inner sleeve of FIG. 4 including tissue received within a lumen of the inner sleeve;

FIG. 7A is a schematic cut-away view of a portion of the inner sleeve of FIG. 4 including an insulative layer;

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FIG. 7B is a schematic cut-away view of a portion of the inner sleeve of FIG. 4 including the insulative layer peeling away from the inner sleeve;

FIG. 8A is a cross-sectional view of the inner sleeve of FIG. 6B taken along line 7A-7A;

FIG. 8B is a cross-sectional view of the inner sleeve of FIG. 6B taken along line 7B-7B;

FIG. 9 is schematic view of a distal portion of another embodiment of the inner sleeve;

FIG. 10A is a cross-sectional view of the inner sleeve of FIG. 9 taken along line 9A-9A;

FIG. 10B is a cross-sectional view of the inner sleeve of FIG. 9 taken along line 9B-9B; and

FIG. 11 is an enlarged sectional view of a working end with an inner sleeve in a fully advanced position illustrating a tissue displacement member.

While the disclosure is amenable to various modifications and alternative forms, specifics thereof have been shown by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to be limited to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosure.

DETAILED DESCRIPTION

For the following defined terms, these definitions shall be applied, unless a different definition is given in the claims or elsewhere in this specification.

All numeric values are herein assumed to be modified by the term “about”, whether or not explicitly indicated. The term “about” generally refers to a range of numbers that one of skill in the art would consider equivalent to the recited value (e.g., having the same function or result). In many instances, the terms “about” may include numbers that are rounded to the nearest significant figure.

The recitation of numerical ranges by endpoints includes all numbers within that range (e.g. 1 to 5 includes 1, 1.5, 2, 2.75, 3, 3.80, 4, and 5).

As used in this specification and the appended claims, the singular forms “a”, “an”, and “the” include plural referents unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

It is noted that references in the specification to “an embodiment”, “some embodiments”, “other embodiments”, etc., indicate that the embodiment described may include one or more particular features, structures, and/or characteristics. However, such recitations do not necessarily mean that all embodiments include the particular features, structures, and/or characteristics. Additionally, when particular features, structures, and/or characteristics are described in connection with one embodiment, it should be understood that such features, structures, and/or characteristics may also be used connection with other embodiments whether or not explicitly described unless clearly stated to the contrary.

The following detailed description should be read with reference to the drawings in which similar elements in different drawings are numbered the same. The drawings, which are not necessarily to scale, depict illustrative embodiments and are not intended to limit the scope of the disclosure.

FIG. 1 illustrates an assembly that comprises an endoscope or hysteroscope 50 used for hysteroscopy together with a tissue-extraction device 100 extending through work-

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ing channel **102** of hysteroscope **50**. Hysteroscope **50** may include handle **104** coupled to elongated shaft **105** having a diameter of 5 mm to 7 mm. Working channel **102** therein may be round, D-shaped or any other suitable shape. Hysteroscope shaft **105** may further be configured with optics channel **106** and one or more fluid inflow/outflow channels **108a**, **108b** (as seen in FIG. 2). Fluid inflow/outflow channels **108a**, **108b** may be in fluid communication with valve-connectors **110a**, **110b** configured for coupling to fluid inflow source **120**, or optionally a negative pressure source **125**. Fluid inflow source **120** may be a component of a fluid management system which may comprises one or more fluid containers and a pump mechanism which pumps fluid through hysteroscope **50** into the uterine cavity. Handle **104** of hysteroscope **50** may include angled extension portion **132** with optics to which videoscopic camera **135** can be operatively coupled. Light source **136** may also be coupled to light coupling **138** on handle **104** of hysteroscope **50**. Working channel **102** of hysteroscope **50** maybe configured for insertion and manipulation of tissue-resecting and extracting device **100**, for example to treat and remove polyp tissue. In some embodiments, hysteroscope shaft **105** may have an axial length of 21 cm, and can comprise a 0° scope, or 15° to 30° scope, for example.

Still referring to FIG. 1, tissue-resecting device **100** may have a highly elongated shaft assembly **140** configured to extend through working channel **102** in hysteroscope **50**. Handle **142** of tissue-resecting device **100** may be adapted for manipulating electrosurgical working end **145** of tissue-resecting device **100**. In use, handle **142** can be manipulated both rotationally and axially, for example, to orient working end **145** to resect targeted polyp tissue. Tissue-resecting device **100** may have one or more subsystems coupled to handle **142** to enable electrosurgical resecting of targeted tissue. For instance, in some embodiments, radiofrequency generator (RF) source **150** and controller **155** may be coupled to at least one RF electrode carried by working end **145**, as described in detail below. In at least some embodiments, electrical cable **156** may be operatively coupled to connector **158** in handle **142**. Electrical cable **156** couples RF source **150** to electrosurgical working end **145**. Exemplary tissue resection devices are described in U.S. Pat. No. 8,512,326, US 2014/0221997, US 2013/0046304, and US 2014/0114300, each of which is herein incorporated by reference in its entirety.

FIG. 1 further illustrates seal housing **162** that carries flexible seal **164** carried by hysteroscope handle **104** for sealing the shaft **140** of tissue-resecting device **100** in working channel **102** to prevent distending fluid from escaping from a uterine cavity. In some embodiments, as shown in FIG. 1, handle **142** of tissue-resecting device **100** may include motor drive **165** for reciprocating, rotating or otherwise moving a resecting component of electrosurgical working end **145**. Handle **142** optionally includes one or more actuator buttons **166** for actuating the tissue-resecting device **100**. In other embodiments, a footswitch can be used to operate tissue-resecting device **100**. In general, a system including at least hysteroscope **50** and tissue-resecting device **100** may include a switch or control mechanism to provide a plurality of reciprocation speeds, for example 1 Hz, 2 Hz, 3 Hz, 4 Hz and up to 8 Hz. The system may further include a mechanism for moving and locking the reciprocating resecting sleeve in a non-extended position, in an extended position, or in an intermediate position. In some embodiments, the system can further include a mechanism for actuating a single reciprocating stroke.

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FIG. 3 illustrates fluid management system **10** that can be used in conjunction with hysteroscope **50** and tissue-resecting device **100** of FIG. 1. Exemplary closed system fluid management systems are described in US 2013/0172805, US 2013/0079702, US 2014/0303551 and US 2015/0119795, each of which is herein incorporated by reference in its entirety. Referring to FIG. 3, in general, fluid management system **10** may comprise fluid source or reservoir **60** containing distention fluid **12**. Controller **25** and two positive displacement (peristaltic) pumps (first infusion pump **40A**, second outflow pump **40B**) may provide fluid inflows and outflows adapted to maintain distension of the uterine cavity. Filter system **80** may also be included for filtering distention fluid **12** that is removed from the uterine cavity **77** and thereafter returned to fluid reservoir **60**. The use of recovered and filtered distention fluid **12** and the replenishment of the volume in fluid reservoir **60** maybe advantageous over open loop systems which do not recover fluid. For instance, closed-loop systems, such as system **10** can effectively measure fluid deficit during a procedure and can provide fluid deficit warnings to insure patient safety. Closed-loop systems may also use only a single bag of distention fluid having a useable volume of about 2500 ml and provide a system lock-out to terminate a procedure after use of a predetermined amount of intravasation of the distention fluid, as determined by measurement of the distention fluid returned to reservoir **60**. Closed-loop systems can also reduce procedure cost by reducing the cost of used distention fluid and fluid disposal costs. Further, closed-loop systems can be set up and operated in a more time-efficient manner, and the systems can be more compact and less expensive than current open loop systems.

As illustrated in FIG. 3, fluid management system **10** can include controller **25**, which can be either independent of tissue-resection device **100** or configured to operate both fluid management system **10** and tissue-resection device **100** where resection device **100** does not include a motor or controller **155**. Controller **25** can be configured to control first and second peristaltic pumps **40A** and **40B** for providing inflows and outflows of distention fluid **12** from reservoir **60** for the purpose of distending uterine cavity **77** and controlling the intra-cavity pressure during various procedures utilizing hysteroscope **50** and/or tissue-resection device **100**.

In some embodiments of FIG. 3, controller **25** may control peristaltic pump **40A** to provide positive pressure at the outflow side **182** of the pump to provide inflows of distention fluid **12** through first flow line or inflow line **45** which is in communication with luer fitting **114a** and fluid flow channel **108a** of hysteroscope **50**. Controller **25** may further control second peristaltic pump **40B** to provide negative pressure at the inflow side **188** of the pump to second flow line or outflow line **35** to assist in providing outflows of distention fluid **12** from the uterine cavity **77**. In operation, second peristaltic pump **40B** may also operate to provide positive pressure on outflow side **190** of pump **40B** in the second outflow line portion **95** to pump outflows of distention fluid **12** through the filter system **80** and back to fluid reservoir **60**.

In some system variations, controller **25** has control algorithms that operate to control pressure in the uterine cavity **77** by pressure signals from a disposable pressure sensor **22** that is coupled to a fitting **114b** of hysteroscope **50** which communicates with flow channel **108b** that extends through hysteroscope shaft **105** to uterine cavity **77**. Pressure sensor **22** can be operatively coupled to controller **25** by cable **20** which sends pressure signals to controller **25**. In

one embodiment, flow channel **108b** has a diameter large enough to allow highly accurate sensing of actual intra-cavity pressure. In other devices, the intra-cavity pressure is typically estimated by various calculations using known flow rates through a pump or remote pressure sensors in the fluid inflow line and/or outflow lines that sometimes rely on back pressure calculations. Such fluid management systems are stand-alone systems that are adapted for use with a variety of hysteroscopes. Most such systems are not able to use a pressure sensor that measures actual intra-cavity pressure. Thus, these other devices and fluid management systems rely on algorithms and calculations to estimate intra-cavity pressure, which are typically less accurate than directly sensing intra-uterine pressure.

Fluid channel or sensor channel **108b** in communication with pressure sensor **22** maybe independent of flow channel **108a** used for inflows of saline into uterine cavity **77**. In the absence of fluid flows in channel **108b**, for example where another channel of hysteroscope **50** or tissue-resecting device **100** is used for fluid outflows, the fluid in the channel **108b** then forms a static column of fluid (air or liquid) that transmits changes in pressure to sensor **22** as the pressure in the uterine cavity changes. In one variation, sensor channel **108b** has a cross-section of at least 1 mm, and fluid pressure within the pressure channel column is equivalent to the pressure in the uterine cavity. Thus, pressure sensor **22** is capable of a direct measurement of pressure within the uterine cavity or other body cavity. In one method, the sensor channel **108b** can be purged of air by opening a valve (not shown) to release air from channel **108b** and sensor **22**.

FIG. 3 schematically illustrates fluid management system **10** in operation in a diagnostic procedure. Uterine cavity **77** is a potential space and needs to be distended to allow for hysteroscopic viewing. A selected pressure can be set in controller **25**, for example via touch screen **205**, which the physician knows from experience is suited for distending cavity **77** and/or for performing the diagnostic procedure. In one variation, the selected pressure can be any pressure between 0 and 150 mm Hg. The first peristaltic pump **40A** may be operated by controller **25** to operate as a variable speed positive displacement pump that is actuated on demand to provide a flow rate from zero up to 1000 ml/min through inflow line **45**. Second peristaltic pump **40B** may be operated at a fixed speed to move the saline distention fluid from uterine cavity **77** through outflow line **35**. In use, controller **25** and a control algorithm can operate pumps **40A** and **40B** at selected matching or non-matching speeds to increase, decrease or maintain the volume of distention fluid **12** in uterine cavity **77**. Thus, by independent control of the pumping rates of first and second positive displacement pumps **40A** and **40B**, a selected set pressure in the body cavity can be achieved and maintained in response to signals of actual intra-cavity pressure provided by pressure sensor **22**.

In FIG. 3, fluid management system **10** is depicted schematically in conjunction with hysteroscope **50**, for example to examine uterine polyp **75**. However, fluid management system **10** may further be used with tissue-resecting device **100** to resect polyp **75**. For example, tissue-resecting device **100** may be inserted through working channel **102** of hysteroscope **50**. In some of these embodiments, outflow line **35** may then be connected to handle **142** of tissue-resecting device **100**, and distention fluid **12** may flow out of uterine cavity **77** through a channel of tissue-resecting device **100** and through outflow line **35**.

Referring to FIGS. 1 and 4, electrosurgical tissue-resecting device **100** includes elongate shaft assembly **140** extend-

ing about longitudinal axis **168** comprising an exterior or first outer sleeve **170** defining passageway or lumen **172**. Lumen **172** may accommodate a second or inner sleeve **175** that can reciprocate (and optionally rotate or oscillate) within lumen **172** to resect tissue. In some embodiments, tissue-receiving window **176** in the outer sleeve **170** has an axial length ranging between about 5 mm to about 10 mm, and in some specific embodiments 8 mm, which may correspond to a size of polyps that tissue-resecting device **100** is designed to remove. In other embodiments, tissue-receiving window **176** may be between about one percent and about three percent of the length of inner sleeve **175** or extraction lumen **160**. Tissue-receiving window **176** may extend in a radial angle about outer sleeve **170** from about 45° to about 210° relative to axis **168** of sleeve **170**. Outer and inner sleeves **170** and **175** can comprise a thin-wall stainless steel material and function as opposing polarity electrodes as will be described in detail below.

FIG. 6A-7B illustrate insulative layers that may be carried by outer and inner sleeves **170** and **175** to limit, control, and/or prevent unwanted electrical current flows between certain portions of sleeve **170**. In some embodiments, outer sleeve **170** may have an O.D. of about 0.143" (3.63 mm) with an I.D. of about 0.133" (3.38 mm). With an inner insulative layer, outer sleeve **170** may have a nominal I.D. of about 0.125" (3.18 mm). In this embodiment, inner sleeve **175** may have an O.D. of about 0.120" (3.05 mm) with an I.D. of about 0.112" (2.84). Inner sleeve **175** with an outer insulative layer may have a nominal O.D. of about 0.123" (3.12) to about 0.124" (3.15 mm) to reciprocate in lumen **172**. In general, insulative layers **200** and **202** may have a thickness between about 0.0005" (0.0127 mm) to about 0.0015" (0.038 mm), and in some specific embodiments about 0.001" (0.025 mm). In other embodiments, outer and or inner sleeves **170** and **175** can be fabricated of metal, plastic, ceramic or a combination thereof. The cross-section of the sleeves can be round, oval or any other suitable shape.

In some embodiments, outer sleeve **170** is made from 304 stainless steel, or other lower cost and lower strength biocompatible steels, and may have an O.D. of about 0.140" (3.56 mm) to about 0.143" (3.63 mm) with a wall thickness of about 0.005" (0.13 mm) to about 0.007" (0.18 mm). In these embodiments, inner sleeve **175** may also be made from 304 stainless steel or other suitable lower cost steels. It can be understood that having the largest possible diameter extraction lumen **160** (FIG. 5) may be advantageous, but the diameter of lumen **160** is limited by the O.D. of the shaft assembly, which in turn is limited by the desired cross section of hysteroscope **50**. To minimize dilation of the patient's cervix, the maximum scope diameter should be about 0.256" (6.5 mm) which generally may allow for a maximum working channel of about 0.150" (3.81 mm). In some example embodiments, the thin wall tubing and insulation layers may be sized to provide an optimized tissue extraction lumen diameter (given the above scope dimensions and limitations above) that is greater than about 0.090" (2.29 mm) or greater than about 0.100" (2.54 mm)—all accommodated in hysteroscope **50** having an O.D. of about 0.256" (6.5 mm).

Thus, in general, tissue resecting device **100** may comprise an elongated assembly comprising concentric outer and inner sleeves extending along an axis, a tissue-receiving window in the outer sleeve and a reciprocating inner sleeve having an extraction lumen **160**. Additionally, the ratio of the diameter of extraction lumen **160** to the outer diameter of outer sleeve **170** is at least about 0.65:1 to about 0.70:1.

In another aspect, the diameter of extraction lumen **160** to the outer diameter of hysteroscope **50** is at least about 0.35:1 to about 0.40:1.

As can be seen in FIG. 4, a distal end of inner sleeve **175** may comprises a first polarity electrode with distal resecting electrode edge **180(+)** about which plasma can be generated. Electrode edge **180(+)** also can be described as an active electrode during tissue resecting since electrode edge **180(+)** then has a substantially smaller surface area than the opposing polarity or return electrode. In some embodiments, as in FIG. 4, the exposed surfaces of outer sleeve **170** may comprise second polarity electrode **185(-)**, which thus can be described as the return electrode since during use electrode **185(-)** has a substantially larger surface area compared to the functionally exposed surface area of the active electrode edge **180(+)**.

As described, inner sleeve or resecting sleeve **175** may have an interior tissue extraction lumen **160** with first and second interior diameters that are adapted to electrosurgically resect tissue volumes rapidly and consistently extract the resected tissue strips through elongated lumen **160** without clogging. Referring now to FIGS. 5 and 6A, it can be seen that inner sleeve **175** may have a first portion **190A** having a first diameter as indicated at A. First portion **190A** may extend from handle **142** (FIG. 1) to distal region **192** of sleeve **175** where tissue extraction lumen **160** transitions to a second portion **190B** with a reduced diameter indicated at B. The diameter of second portion **190B** is defined by electrode sleeve element **195** that provides resecting electrode edge **180**. The axial length C of the second portion **190B** can range from about 1 mm to about 15 mm. In some embodiments, the first diameter A is about 0.106" (2.69 mm) and the second reduced diameter B is about 0.095" (2.41 mm) and has an axial length of about 2 mm. The cross-sectional area of second portion **190B** may be less than 95% of cross-sectional area of first portion **190A**, or less than 90% of the cross-sectional area of first portion **190A**, or 85%, or 80% in other embodiments. As shown in FIG. 5, inner sleeve **175** can be an electrically conductive stainless steel, and second portion **190B** can also comprise stainless steel electrode sleeve element **195** that is welded in place by weld **196** (FIG. 6A). In other alternative embodiments, inner sleeve **175** and electrode sleeve element **195** can comprise a tungsten tube that can be press fit into distal end **198** of inner sleeve **175**.

FIGS. 5 and 6A further illustrate the interfacing insulation layers **202** and **204** that may be carried by first and second sleeves **170**, **175**, respectively. In FIG. 6A, outer sleeve **170** is lined with a thin-wall insulative material **200**, such as perfluoroalkoxy alkane (PFA), or other polymeric materials. Similarly, inner sleeve **175** may have an exterior insulative layer **202**. These insulative layers can be lubricious as well as electrically insulative to reduce friction during reciprocation of inner sleeve **175**. Insulative layers **200** and **202** can comprise a lubricious, hydrophobic or hydrophilic polymeric material. For example, the material can comprise a bio-compatible material such as TEFLON®, polytetrafluoroethylene (PTFE), fluorinated ethylenepropylene (FEP), polyethylene, polyamide, ECTFE (Ethylenechlorotrifluoroethylene), ETFE, PVDF, polyvinyl chloride, silicone, or the like.

Turning to FIG. 6B, another variation of inner sleeve **175** is illustrated in a schematic view together with a tissue volume being resected with plasma electrode edge **180**. In this embodiment, as in other embodiments, an RF source

The plasma generated at electrode edge **180** can resect and ablate a path P in tissue **220**, as tissue resecting device **100** is movable relative to tissue **220** or inner sleeve **175** is moved relative to outer sleeve **170**, and is suited for resecting uterine polyp tissue and other abnormal uterine tissue. As seen in FIG. 6B, the distal portion of resecting sleeve **175** may include ceramic collar **222** adjacent to electrode sleeve element **195**. In these embodiments, insulative layer **202** may extend over inner sleeve **175**, but may not contact electrode sleeve element **195**. In these embodiments, insulative layer **202** may instead be attached to ceramic collar **222**. Ceramic collar **222** collar may function to confine plasma formation about distal electrode edge **180** and help to prevent plasma from contacting and damaging polymer insulative layer **202** on resecting sleeve **175** during operation.

However, in other embodiments, as depicted in FIG. 7A, tissue-resecting device **100** may not include ceramic collar **222**. In these embodiments, insulative layer **202** may extend distally beyond a distal end of inner sleeve **175** and cover at least a portion of electrode sleeve element **195**. For instance, insulative layer **202** may be bonded directly to electrode sleeve element **195**. Without a ceramic collar, the plasma generated at electrode edge **180(+)** during resection may wear down insulative layer **202** more quickly than if a ceramic collar had been included between electrode sleeve element **195** and insulative layer **202**. In some additional embodiments, insulative layer **202** may be comprised of a material that may be less wear-resistant or degrade relatively more easily than other materials. For instance, insulative layer **202** may comprise FEP, as opposed to a polyester material.

In some embodiments, tissue-resecting device **100** may be configured to be used only for particular procedures, such as for resecting uterine polyps, or used for a particular number of procedures. For instance, uterine polyps may be generally less fibrous or mechanically rigid than uterine fibroids. Accordingly, the materials of tissue-resecting device **100** configured for uterine polyp resection may not need to be as highly wear-resistant or stand up to a higher level of forces that may be present during resection of uterine fibroids. Utilizing less-wear resistant materials and/or weaker materials may allow tissue-resecting device to be built for a lower cost than devices configured for resection of uterine fibroids. For example, inner and outer sleeves **170**, **175** may be comprised of 304 stainless steel or another lower strength bio-compatible stainless steel. Additionally, at least insulative layer **202** may be comprised of FEP as opposed to more durable materials, such as polyesters or other polymers.

In these embodiments, insulative layer **202** may be configured to peel back from electrode sleeve element **195** and/or inner sleeve **175**, as depicted in FIG. 7B, after a duration of time of using the device **100**, a particular number of activations of tissue-resection device **100** or after a particular total length of activation. In additional or alternative embodiments, insulative layer **202** may be configured to peel back from electrode sleeve element **195** and/or inner sleeve **175** when tissue-resection device **100** is used for procedures imparting a greater amount of force on tissue-resection device **100** than during resection of uterine polyps, such as where tissue-resection device **100** is used to resect uterine fibroids or other abnormal tissue that is more fibrous than uterine polyps. Activation of tissue-resection device **100** may include providing RF energy through electrode sleeve element **195** and/or reciprocation of inner sleeve **175** relative to outer sleeve **170**. Once insulative layer **202** peels back or otherwise becomes detached from inner sleeve **175**,

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the normal flow pathway of the RF energy may change, resulting in tissue-resecting device **100** becoming non-operational. In this manner, tissue-resection device **100** may be configured to fail or stop working under conditions different from those for which tissue-resection device **100** was designed. For example, the portion of insulative layer **202** exposed in the window **176** of outer sleeve **170** as the inner sleeve **175** moves to the distally extended position (distal or window-closed position) may wear or become delaminated from the inner sleeve **175** through repeated frictional contact with tissue during reciprocation of the inner sleeve **175** relative to the outer sleeve **170** (thus reducing the degree of contact between the insulative layer **202** and the inner sleeve **175**), which may expose a portion of the inner sleeve **175** causing a modified or altered electrical pathway between a now exposed electrically conductive portion of the inner sleeve **175** and the exposed electrically conductive portion of the outer sleeve **170** serving as the return electrode. Such modification of the electrical pathway may cause an electrical short or impedance change, making the device **100** non-operational.

Referring back to FIG. 6B, in some aspects, the path P formed in tissue **220** with the plasma at electrode edge **180** may provide a path P having an ablated width indicated at W, where such path width W is substantially created due to tissue vaporization. This removal and vaporization of tissue in path P is different than the effect of cutting similar tissue with a sharp blade edge, as in various prior art devices. A sharp blade edge can divide tissue (without cauterization) but applies mechanical force to the tissue and may prevent a large cross section slug of tissue from being cut. In contrast, the plasma at the electrode edge **180** can vaporize a path P in tissue without applying any substantial force on the tissue to thus resect larger cross-sections of strips of tissue. Further, the plasma resecting effect reduces the cross section of tissue strip **225** received in the tissue-extraction lumen of second portion **190B**. FIG. 6B depicts tissue strip **225** entering the lumen of second portion **190B** which has a smaller cross-section than the lumen of second portion **190B** due to the vaporization of tissue. Further, the cross section of tissue **225** as it enters the larger cross-section lumen of first portion **190A** results in even greater free space **196** around the tissue strip **225**. Thus, the resection of tissue with plasma electrode edge **180**, together with the lumen transition from the smaller cross-section of second portion **190B** to the larger cross-section of first portion **190A** of tissue-extraction lumen **160** can significantly reduce or eliminate the potential for successive resected tissue strips **225** to clog lumen **160**. Prior art resection devices with smaller diameter tissue-extraction lumens typically have problems with tissue clogging.

In other aspects where a system includes a negative pressure source coupled to the proximal end of tissue-extraction lumen **160**, the negative pressure source may also assist in aspirating and moving tissue strips **225** in the proximal direction to a collection reservoir (not shown) outside handle **142** of the device.

FIGS. 8A-8B illustrate the change in lumen diameter **160** of resecting sleeve **175'** of FIG. 6B. FIG. 9 illustrates the distal end of a variation of resecting sleeve **175'** which is configured with electrode sleeve element **195'** that is partially tubular in contrast to the previously described tubular electrode sleeve element **195** (FIGS. 5 and 6A). FIGS. 10A-10B again illustrate the change in cross-section of tissue-extraction lumen **160** between second portion **190B'** having a reduced cross-section and first portion **190A'** having an increased cross-section region **190A'** in relation to

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resecting sleeve **175'** of FIG. 9. Thus, the functionality remains the same whether electrode sleeve element **195'** is tubular or partly tubular. In FIG. 9, ceramic collar **222'** is shown, in one variation, as extending only partially around sleeve **175'** to cooperate with the radial angle of electrode sleeve element **195'**. Further, the variation of FIG. 9 illustrates that ceramic collar **222'** has a larger outside diameter than insulative layer **202**. Thus, friction may be reduced since the short axial length of ceramic collar **222'** interfaces and slides against interfacing insulative layer **200** about the inner surface of lumen **172** of outer sleeve **170**. However, in other embodiments, resecting sleeve **175'** may not include ceramic collar **222'**, as described with respect to sleeve **175**.

In some aspects, the axial length of tissue-extraction lumen **160** may range between from about 17.7" (450 mm) to about 21.7" (550 mm) for access to a uterine cavity. In some embodiments, shaft assembly **140** of tissue-resecting device **100** may be about 35 cm in length. However, in other embodiments, shaft assembly **140** include tissue-extraction lumen **160** that is at least about 10 cm, about 20 cm, about 30 cm, or about 40 cm in length.

Now referring to FIG. 4 and FIG. 11, one aspect of the disclosure comprises a "tissue displacement" mechanism that is configured to displace and move tissue strips **225** in the proximal direction in lumen **160** of inner sleeve **175** to ensure that tissue **225** does not clog lumen **160**. As can be seen in FIG. 4 and FIG. 11, one tissue displacement mechanism comprises projecting element **230** that extends proximally from distal tip or body **232** that is fixedly attached to outer sleeve **170**. Projecting element **230** may extend proximally along central axis **168** in a distal chamber **240** defined by outer sleeve **170** and the interior surface of distal tip **232**. In some embodiments, as depicted in FIGS. 4 and 11, shaft-like projecting element **230** thus may function as a plunger or pushing member and can push captured tissue strip **225** in the proximal direction from the lumen of second portion **190B** of inner sleeve **175** as sleeve **175** moves to its fully advanced or extended position (FIG. 11). For this reason, the length D of projecting element **230** may be at least as great as the axial length E of the second portion **190B** of inner sleeve **175**. Further, as depicted in FIG. 11, the stroke Y of inner sleeve **175** extends at least about 3 mm, 4 mm, or 5 mm distally beyond the distal edge of window **290**. In another aspect, the stroke Y of inner sleeve **175** may be at least 5% or 10% of the total stroke of inner sleeve **175** (stroke X+stroke Y in FIG. 11).

In general, displacement feature or projecting element **230** may have a maximum cross-section that extends substantially across a cross-section of extraction lumen **160**. In some variations, displacement feature **230** may have a cross-sectional area that substantially occupies the cross-sectional area of second portion **190B** of inner sleeve **175**. FIGS. 4 and 11 illustrate projecting element **230** as cylindrical. However, in other embodiments projecting element **230** may be shaped differently. For instance, projecting element **230** may have a symmetric shape relative to a central axis of extraction lumen **160**, and may be star-shaped or fluted with ribs and channels to allow distension fluid to flow therethrough. In other embodiments, projecting element **230** can have an asymmetric cross sectional shape with any number or flutes, grooves, lumens or bore extending about its axis. In at least some embodiments, projecting element **230** may be comprised of a dielectric material such as a ceramic or polymer.

In some aspects, the tissue resecting device may comprise an elongated assembly comprising concentric outer and inner sleeves, with a tissue-receiving window in the outer

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sleeve open to an interior lumen with a distal lumen portion extending distal to the window. The inner sleeve may further be configured with a first axially-extending channel having a greater cross-sectional area and a second axially-extending channel portion having a second smaller cross-sectional area and wherein the ratio of lengths of the distal lumen portion relative to the first channel at least 1:1. In some embodiments, the device may be configured with a length of the distal lumen portion that is at least 5 mm. In these embodiments, the length of the first axially-extending channel may be at least 5 mm. In other embodiments, the ratio of lengths of the distal lumen portion relative to the diameter of the interior lumen is at least 1:1. In still other embodiments, the ratio is at least 1.5:1. In these embodiments, the length of the distal lumen portion may be at least 5 mm. In other variations, the diameter of the interior lumen is less than 5 mm.

In other aspects, a tissue resecting device may comprise a handle coupled to an elongated tubular assembly comprising outer and inner concentric sleeves and a tissue-receiving window in the outer sleeve communicating with an interior passage-way extending through the assembly. In some of these embodiments, a distal edge of the window may be spaced at least 4 mm, 6 mm, 8 mm, or 10 mm from the distal end of the interior passageway. In these variations, the mean cross-section of the passageway may be less than 5 mm, 4 mm, or 3 mm.

Some embodiments of a tissue resecting device comprise a handle coupled to an axially-extending shaft assembly defining a tissue-receiving window communicating with an interior extraction lumen for extracting tissue. The shaft assembly may comprise axially-extending first and second elements with at least one element axially moveable relative to the other element between a first position and a second position, and a displacement feature configured to displace resected tissue from the extraction lumen. In these embodiments, the first position may comprise an open-window configuration for receiving tissue therein and the second position is a closed-window configuration. The movement of the elements from the first position toward the second position resects tissue with an edge of one of the elements. The edge may comprise an RF electrode edge. The displacement feature (FIG. 4 and FIG. 11) or projecting element 230 can be coupled to the first element and can project axially relative to an axis of the extraction lumen. These embodiments may be configured with an extraction lumen having first and second cross-sectional areas, wherein a distal region of the extraction lumen has a first lesser cross-sectional area and a medial portion of the extraction lumen has a second greater cross-sectional area. In some variations, the distal region of the extraction lumen may have the first cross-sectional area extends axially at least 2 mm, 4 mm, 6 mm, or 8 mm. In other variations, the displacement feature may be configured to extend axially into the extraction lumen in the second closed-window configuration at least 2 mm, 4 mm, 6 mm, or 8 mm.

Some methods of resecting tissue may comprise resecting tissue with a reciprocating sleeve having an extending stroke and a retracting stroke within an outer sleeve, wherein the extending stroke resects and captures tissue received by a tissue-receiving window in the outer sleeve. The method may further comprise pushing the captured tissue in the proximal direction in the inner sleeve with a displacement member when the inner sleeve is in a transition range in which the inner sleeve transitions from the extending stroke to the retracting stroke. Further, the displacement member may be configured to push the captured tissue at least in part from a second portion of the inner sleeve having a smaller

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cross-section lumen to a first portion of the inner sleeve having a larger cross-section lumen. Thereafter, the negative pressure source can more effectively extract and aspirate the tissue from the lumen.

In some variations, the resecting step can include applying RF current to generate plasma at an electrode edge 180 on inner sleeve 175 and further comprising the step of terminating RF current at the distal end of the first resecting stroke. Alternatively, the system and controller 155 can terminate RF current during the second resecting stroke. Alternatively, the controller 155 can terminate RF current during the retracting stroke.

In a further variation, the controller can apply RF current to the electrodes during at least a portion of the retracting stroke to thereby cauterize adjacent tissue. The cautery effect can be provided during the retracting stroke at the same operational parameters as used during the first resecting stroke, or at different operational RF parameters than used during the first resecting stroke.

Those skilled in the art will recognize that the present disclosure may be manifested in a variety of forms other than the specific embodiments described and contemplated herein. Specifically, the various features described with respect to the various embodiments and figures should not be construed to be applicable to only those embodiments and/or figures. Rather, each described feature may be combined with any other feature in various contemplated embodiments, either with or without any of the other features described in conjunction with those features. Accordingly, departure in form and detail may be made without departing from the scope of the present disclosure as described in the appended claims.

What is claimed:

1. A tissue resecting device comprising:

an elongated structure having a longitudinal axis, the elongated structure comprising an outer sleeve with a distal window configured to receive tissue and an inner sleeve configured to move between a proximal position and a distal position relative to the window;

an electrode element coupled to the inner sleeve and extending distally of the inner sleeve, wherein the electrode element is distinct from the inner sleeve; and an insulative layer disposed on an outer surface of the inner sleeve and extending uninterrupted onto an outer surface of the electrode element, wherein the outer surface of the electrode element extends distal of the insulative layer;

wherein the electrode element serves as an active electrode having a first polarity and the outer sleeve serves as a return electrode having a second polarity opposite the first polarity.

2. The tissue resecting device of claim 1, wherein the insulative layer is bonded directly to the outer surface of the electrode element.

3. The tissue resecting device of claim 1, wherein the insulative layer is configured to wear away or delaminate from the outer surface of the electrode element after a predetermined duration of time of using the tissue resecting device.

4. The tissue resecting device of claim 1, wherein the insulative layer is configured to wear away or delaminate from the outer surface of the electrode element after a predetermined number of activations of the tissue resecting device.

5. The tissue resecting device of claim 4, wherein activation of the tissue resecting device comprises providing RF energy to the electrode element.

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6. The tissue resecting device of claim 4, wherein activation of the tissue resecting device comprises axially reciprocating the inner sleeve relative to the outer sleeve.

7. The tissue resecting device of claim 1, wherein the insulative layer is configured to wear away or delaminate from the outer surface of the electrode element after a predetermined total activation time of the tissue resecting device.

8. The tissue resecting device of claim 7, wherein activation of the tissue resecting device comprises providing RF energy to the electrode element.

9. The tissue resecting device of claim 7, wherein activation of the tissue resecting device comprises axially reciprocating the inner sleeve relative to the outer sleeve.

10. A medical device system for resecting tissue, comprising:

an elongated structure having a longitudinal axis, the elongated structure comprising an outer sleeve with a distal window configured to receive tissue and an inner sleeve configured to move between a proximal position and a distal position relative to the window;

an electrode element coupled to the inner sleeve and extending distally of the inner sleeve, wherein the electrode element is distinct from the inner sleeve; and an insulative layer disposed on an outer surface of the inner sleeve and extending uninterrupted onto an outer surface of the electrode element, wherein the outer surface of the electrode element extends distal of the insulative layer; and

an RF generator for delivering energy through the electrode element to resect tissue;

wherein the electrode element serves as an active electrode having a first polarity and the outer sleeve serves as a return electrode having a second polarity opposite the first polarity.

11. The medical device system of claim 10, wherein the insulative layer is configured to wear away or delaminate from the outer surface of the inner sleeve after a predetermined duration of time of using the medical device system.

12. The medical device system of claim 10, wherein the insulative layer is configured to wear away or delaminate from the outer surface of the inner sleeve after a predetermined number of activations of the medical device system.

13. The medical device system of claim 12, wherein activation of the medical device system comprises providing RF energy to the electrode element.

14. The medical device system of claim 12, wherein activation of the medical device system comprises axially reciprocating the inner sleeve relative to the outer sleeve.

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15. The medical device system of claim 10, wherein the insulative layer is configured to wear away or delaminate from the outer surface of the inner sleeve after a predetermined total activation time of the medical device system.

16. The medical device system of claim 15, wherein activation of the medical device system comprises providing RF energy to the electrode element.

17. The medical device system of claim 15, wherein activation of the medical device system comprises axially reciprocating the inner sleeve relative to the outer sleeve.

18. A method of resecting tissue, comprising: engaging a tissue resecting device with tissue, wherein the tissue resecting device comprises:

an elongated structure having a longitudinal axis, the elongated structure comprising an outer sleeve with a distal window configured to receive tissue and an inner sleeve configured to move between a proximal position and a distal position relative to the window; a tubular electrode element coupled to the inner sleeve and extending distally of the inner sleeve to an open distal end, wherein the electrode element is distinct from the inner sleeve; and

an insulative layer disposed on an outer surface of the inner sleeve and extending uninterrupted onto an outer surface of the electrode element, wherein the outer surface of the electrode element extends distal of the insulative layer;

supplying RF energy along an electrical pathway from the electrode element to the outer sleeve as the inner sleeve is axially reciprocated relative to the outer sleeve to generate plasma at a distal edge of the electrode element to resect the tissue received within the distal window.

19. The method of claim 18, wherein the insulative layer is configured to peel back from the outer surface of the inner sleeve after a predetermined number of activations of the tissue resecting device such that when a portion of the outer surface of the inner sleeve is exposed, the electrical pathway from the electrode element to the outer sleeve is altered, thereby rendering the tissue resecting device non-operational.

20. The method of claim 18, wherein the insulative layer is configured to wear away or delaminate from the outer surface of the inner sleeve after a predetermined total activation time of the tissue resecting device such that when a portion of the outer surface of the inner sleeve is exposed, the electrical pathway from the electrode element to the outer sleeve is altered, thereby rendering the tissue resecting device non-operational.

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